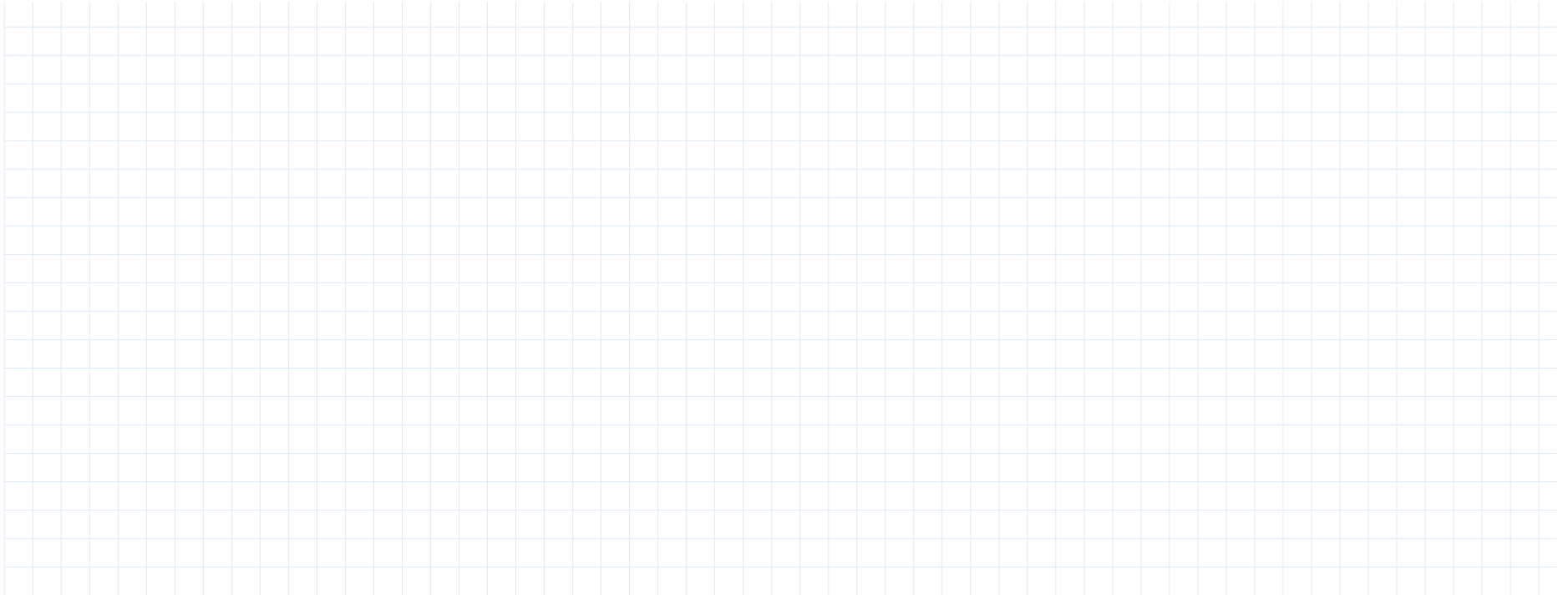


Exponential Functions: The Variable Is in the Exponent

An exponential function has the form $f(x) = Ca^x$, where $a > 0$, $a \neq 1$, and $C \neq 0$. For $C > 0$, $a > 1$ gives growth; $0 < a < 1$ gives decay. The initial value is $f(0) = C$.

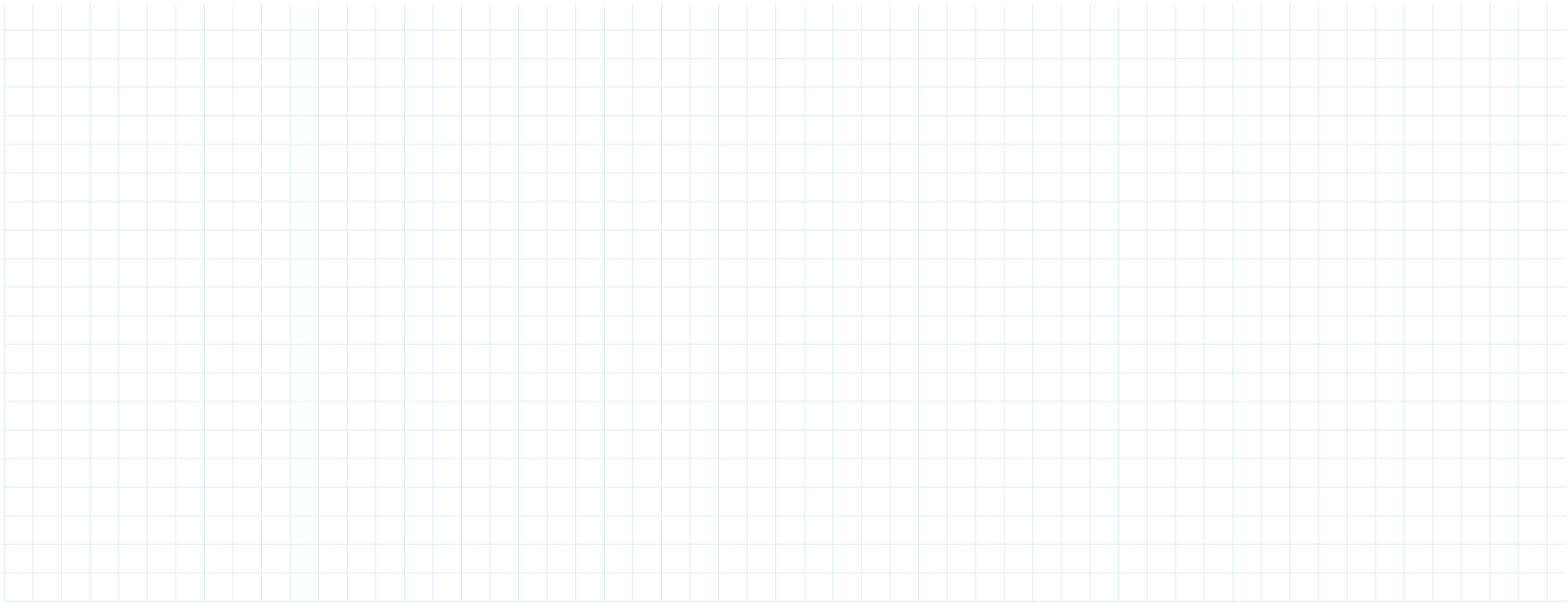
$$f(x) = 2^x, \quad g(x) = \left(\frac{1}{2}\right)^x.$$

Evaluate each at $x = -2, -1, 0, 1, 2$. Identify growth or decay and the y -intercept.



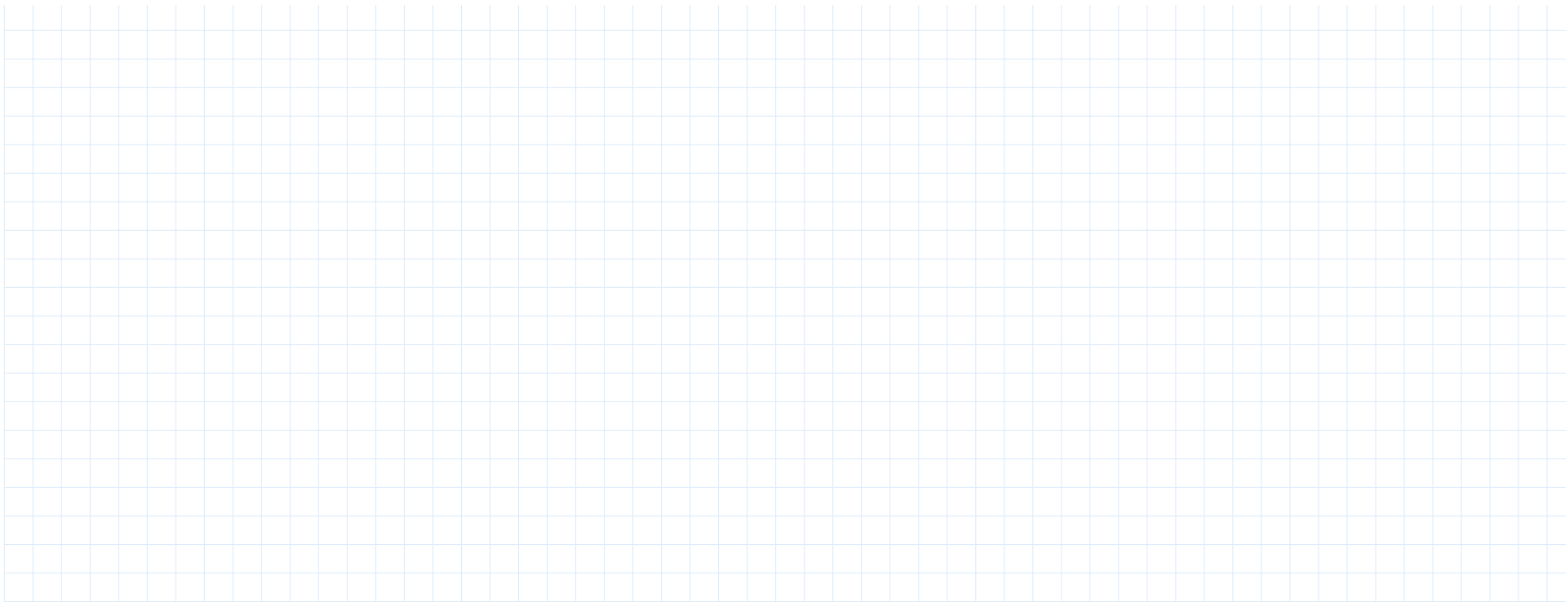
Sketch Growth and Decay

Use the table from the previous page to sketch $f(x) = 2^x$ and $g(x) = \left(\frac{1}{2}\right)^x$. For each function state the domain, range, horizontal asymptote, and whether it has an x -intercept.



A Shifted Exponential Graph

Sketch $h(x) = 3^x - 4$. State the domain, range, horizontal asymptote, and y-intercept. Plot at least three useful points.



Solve by Using the Same Base

For $a > 0$ and $a \neq 1$, the equation $a^u = a^v$ implies $u = v$. Solve by rewriting both sides with the same base:

(a) $2^{3x} = 16$

(b) $3^{2x-1} = 9.$

Check your answers in the original equations.



The Natural Base and a Quadratic Exponent

The constant $e \approx 2.71828$ is a common exponential base; it is not a variable. Solve and check:

(a) $e^{2x} = e^{x+3}$ (b) $e^{x^2} = e^{3x+4}$.

The same-base rule applies to e just as it applies to 2 or 3.

